

Stage 1 (invertible blocks only): $\mathcal{L} = \mathcal{L}_{\text{fwd}} + \mathcal{L}_{\text{inv}} + \mathcal{L}_{\text{dir}, a} + \mathcal{L}_{\text{dir}, y}$

Stage 2 (VAE only): $\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}, \quad \beta = 0.05$

Stage 3 (all parts together): $\mathcal{L} = \mathcal{L}_{\text{fwd}} + \mathcal{L}_{\text{inv}} + \mathcal{L}_{\text{dir}, a} + \mathcal{L}_{\text{dir}, y} + \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}$

$\mathcal{L}_{\text{fwd}}$: ERP predicted from the design, against the true ERP

$\mathcal{L}_{\text{inv}}$: design estimated from the ERP, against the true design

$\mathcal{L}_{\text{dir}, a}$: design read out directly from the design encoding (blocks skipped), against the true design

$\mathcal{L}_{\text{dir}, y}$: ERP read out directly from the ERP encoding (blocks skipped), against the true ERP

$\mathcal{L}_{\text{recon}}$: design rebuilt by the VAE, against the true design

$\mathcal{L}_{\text{KL}}$: keeps the VAE latent distribution close to a standard normal distribution $\mathcal{N}(0, 1)$

All other terms are mean squared errors.